

Learning target DF3, version 1

Compute basic derivatives using algebraic rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum/difference, etc.) you are using in your work.

1. $f(x) = 2 \cos(x) + 4e^x$

2. $h(t) = \sqrt[3]{t^4} + \frac{4}{t^4}$ (Don't use the quotient rule or the chain rule.)

3. $g(w) = 6w^3 - 2w^2 - 9w - 2$

4. $C(t) = 64(e^7 + 7^e) - \frac{\sqrt[27]{193262}}{16 - \pi^2} + \frac{67}{8675309}$ (Hint: Look carefully!)

Learning target DF3, version 2

Compute basic derivatives using algebraic rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum/difference, etc.) you are using in your work.

1. $g(t) = \sqrt[4]{t^3} + \frac{7}{t^4}$ (Don't use the quotient rule or the chain rule.)

2. $f(x) = -x^5 - 4x^4 + 5x - 4$

3. $h(y) = -4 \ln(y) + 3 \sin(y)$

Learning target DF3, version 3

Compute basic derivatives using algebraic rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum/difference, etc.) you are using in your work.

1. $f(y) = -2e^y - 4 \sin(y)$

2. $h(x) = 9x^5 - x^4 - x^3 - 6$

3. $g(t) = \sqrt[3]{t^7} + \frac{2}{t^2}$ (Don't use the quotient rule or the chain rule.)

Learning target DF3, version 4

Compute basic derivatives using algebraic rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum/difference, etc.) you are using in your work.

1. $g(t) = -5t^4 - 9t^3 - 6t^2 - 6$

2. $f(w) = \sqrt[3]{w^5} + \frac{4}{w^5}$ (Don't use the quotient rule or the chain rule.)

3. $h(y) = 2e^y + 2 \sin(y)$